ICS 483 Assignment 2

Shea Stevens

October 21, 2026

1 Theory Questions

1.1

In order to determine the shift in intensity, we compare a window with shifted copies, given by the function $E(u,v) = \sum_{x,y}[I(x+u, y+v) - I(x,y)]^2$. When a flat region is observed, the change between these two windows is very small in all directions. In the case of an edge, we observe a large intensity change when shifting over the edge, and little to no change when shifting along the edge. So, in order to detect a corner, we want to observe many of these edges at a given point in different orientations. It is when these multiple orientation intersect in a single window, that we observe a large E in all directions and consider it a corner.

1.2

$M = \begin{bmatrix} S_{xx} & S_{xy} \\ S_{xy} & S_{yy} \end{bmatrix}$

When we observe a second moment matrix, we derive eigenvalues (λ_1, λ_2) and eigenvectors $M\mathbf{v}_1 = \lambda_1\mathbf{v}_1$. Because we want to see shifts in multiple directions, we categorize flat, corner, and edge based on the two eigenvalues and their relationship to one another. When a surface is flat, we see the both λ_1 and λ_2 are both small. When one is sizably larger than the other ($\lambda_1 >> \lambda_2$) or ($\lambda_1 << \lambda_2$) it is determined as an edge. And finally, when both values are large and in similar size, it is considered a corner. This is because we want to see significant changes in intensity in multiple directions, as previously described in question 1.

1.3

It is impossible to discern whether a single point's gradient lies on a corner or an edge. We can determine the direction and magnitude of the gradient, but without more information from surrounding pixels, it is too difficult to tell whether that magnitude and direction lie on a straight line or corner. As we previously established, the eigenvalues must be equally large in order to constitute a corner. This eigenvalue is decided by the summation of multiple points within the window. So, only when this window contains multiple gradients pointing in different directions can we conclude that it contains a corner.

1.4

Non-maximum suppression's main use is to differentiate between local maxima and non-maxima. Non-maximum suppresion finds local maxima within a window so that, when compared to a threhold that reduces image noise, decreases the number of detected points within said window. We do this to to get one localized corner, which makes tasks like feature matching or stitching more feasible.

2 Implementation Discussion

2.1 Implementation



Figure 1: original image

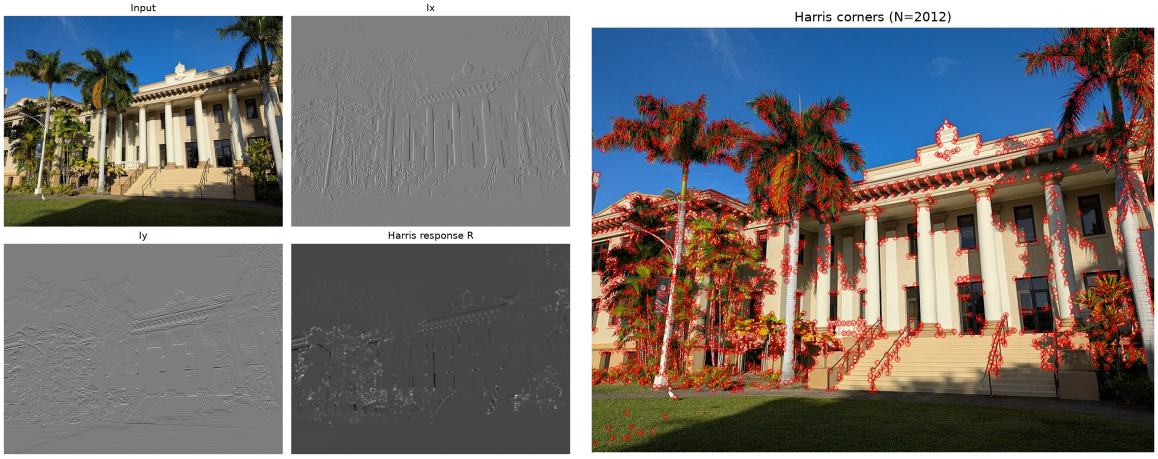


Figure 2: Intermediary edge detection test

Harris corner detection works best on the buildings geometric features. The sharp corners of the building create sharp intensity changes in multiple directions that provide the model with ideal circumstances to detect intensity shifts in multiple directions. Harris corner struggles in three key features: shadows, texture, and color boundaries with similar intensity.

Shadows could be considered an edge case. When the image is considered its own planar space, it would make sense that a shadow indicates a change in intensity. When combined with 'real' edges like the building, it creates the illusion of a corner. If we tried to apply this in conjunction with semantic recognition, the boundaries would be inaccurate. We might want to consider illumination normalization to compensate.

Texture is another struggle point. It's most visible in the grass, where the small blades create high noise, which results in random points scattered amongst the green. We see this also in the palm fronds where blades produce 'corners' as they overlap.

Finally, color boundaries with similar intensity struggle. Because we used intensity shifts to define a corner, different colors with similar intensity will result in no corner being found. It's prominent on the top of the building, where the corners most brightly lit are not detected, but shadows cast on the crest do detect corners.

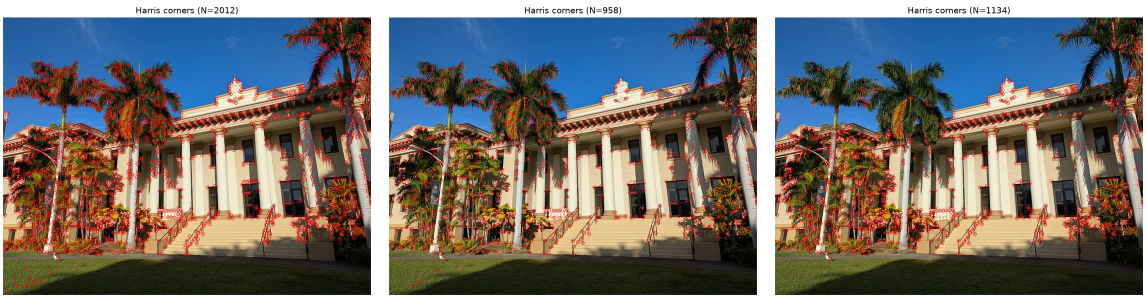


Figure 3: Difference in presets — $\sigma = 1.5$ — $\sigma = 3$ — $thresholdRel = 0.03$

Each of the four presets affect our algorithm in specific ways. Sigma defines the standard deviation of the Gaussian. As σ increases, the Gaussian widens, increasing the window that is weighted. We apply a gaussian to the secondary moment matrix because we a single pixel cannot describe a corner; we need its relationship to other points in the window.

ThresholdRel is one half of the corner decision. By enforcing a threshold, you enforce a minimum intensity change with the form $R > thresholdRel * max(R)$. When thresholdRel increases, stronger intensity changes in multiple directions are required to satisfy the algorithm, as shown by image 1 and 3 in figure 3. nmsSize is the other half, which determines the size of the local maxima window. Because only one point is chosen within this window, increasing this value spaces out the found corners in the final Harris response.



Figure 4: $k = 0.24, k = 0.25$

k is a constant that is inside R. This value determines how much edges are penalized relative to corner. The determinant ($det(M)$) is large only when both eigenvalues are large. The trace ($tr(M)^2$) is large whenever either eigenvalue is large. Our Harris corner detection algorithm uses the following equation to determine a response.

$$R = det(M) - k\ tr(M)^2 = \lambda_1\lambda_2 - k(\lambda_1 + \lambda_2)^2$$

In order for a corner to succeed, we must satisfy two requirements:

$$R > 0 \quad \& \quad \frac{\lambda_1\lambda_2}{(\lambda_1 + \lambda_2)^2} > k$$

So, with equal and large lambda, the highest number the above fraction can equal is 0.25. This matches the outcome shown in figure 4, where $k >= 0.25$ results in no point passing the detection algorithm.

2.2 AI Usage

AI was used along the course of this assignment primarily to understand the underlying concepts required to complete the assignment. It was used in conjunction with the agents.md file that was recommended by Professor Chen, which gates direct terminal code running and did not give me pseudocode. All code submitted and this writeup are entirely hand written by me, Shea Stevens. An

attachment of a concatenated log, which highlights the information given by the model, can be found both as it's own md file, and in the zip file provided for this assignment.